Radmehr Karimian

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Education

ETH Zurich Starting Sep. 2026
Exchange Research Student *Department of Computer Science (D-INFK)*

University of Geneva Sep. 2025 - Present
Master of Science in Statistics

Sharif University of Technology Sep. 2019 - Jul. 2024
Bachelor of Science in Electrical Engineering

Helli 1 High School Aug. 2019
High School Diploma in Mathematics and Physics

Research Interests

My research lies at the intersection of statistical machine learning, foundation and generative models, and AI for scientific discovery. I study the theoretical and empirical behavior of diffusion models, robust and semi-supervised learning, and agentic AI systems for scientific workflows. My broader interests include self-supervised representation learning, reliable and explainable AI, high-dimensional statistics, information theory, machine unlearning, and privacy-preserving learning.

Publications

2026. **Scalable LAD Regression via Statistical-Scale Geometry**.
Available under request, 2026.
R. Karimian*

2026. **Spectral Effective Dimension Controls Diffusion Model Convergence**.
Under Review, 2026.
R.Karimian*, A.Mohades, Johannes Lederer
arXiv

2026. **f-FUM: Federated Unlearning via min–max and f-divergence**.
Under Review, 2026.
R. Karimian*, A. Bagheri*, Meghdad Kurmanji, Nicholas D. Lane, G. Aminian.
arXiv

2025. **f-SCRUB: Unbounded Machine Unlearning with f-divergence**.
DataFM Workshop at ICLR 2025, 2025.
R. Karimian*, A. Bagheri*, G. Aminian
OpenReview

2024. **Robust Semi-supervised Learning via f-Divergence and α-Rényi Divergence**.
In Proceedings of the IEEE International Symposium on Information Theory (ISIT), 2024.
G. Aminian, **R. Karimian***, A. Bagheri*, M. Jafari*.
arXiv

2023. **Clinical IoT in Practice: A Novel Design and Implementation of a Multi-functional Digital Stethoscope for Remote Health Monitoring**.
In TechXiv, 2023.
arXiv

Research Experience and Ongoing Manuscripts

2026. **Autonomous Photonic Integrated Circuit (PIC) Design & Benchmarking.**
EPFL - LWE lab
Engineered agentic frameworks for automated photonic design and simulation, benchmarking frontier AI systems on complex physical-science workflows via PhySciBench. Evaluated long-horizon reasoning, numerical accuracy, and multi-tool orchestration across state-of-the-art models.

2025. **Gradient Descent Methods for Nonsmooth Robust Estimations**.
Columbia University — Supervisor: Prof. Marco Avella Medina
We consider Least Absolute Deviation (LAD) as a robust estimator that is highly valued in financial datasets. However, due to its nonsmooth nature, it requires significant computation time in large-scale inference settings, limited by convex optimization methods such as Interior Point Methods (IPM). Despite these challenges, we demonstrate that linear and fast convergence can be achieved through backtracking, yielding a statistically valid solution, w.h.p.

2025. **Optimal Batch Size under Local Differential Privacy**.
Under preparation, 2025.
Columbia University — Supervisor: Prof. Marco Avella Medina
Investigates optimal batch-size selection under local differential privacy, extending the framework of Duchi, Jordan, and Wainwright to improve computational efficiency while preserving statistical guarantees.

Awards

Best Students Award Sep. 2020
Recognized as one of the top students of the academic year.

Bachelor Thesis Award Oct. 2023
Thesis selected among the top 15% of submissions.

Best Bachelor Thesis Award Oct. 2023
Ranked second overall among shortlisted projects and first in the Information and Communications category

IMS Student Puzzle Winner Dec. 2024
Puzzle 53 Winner

Selected Courses with Project

⋆ **Theoretical Statistics** | Unige
⋆ **Time Series Analysis** | Unige
⋆ **High Dimensional Probability** | SUT
- Adversarial attacks Report Slides

⋆ **Info. Theory, Stat. & Learning** | SUT
- Imbalance & robust semi-sup. Report Slides

⋆ **Deep Learning** | SUT
⋆ **Information Theory** | SUT
Convex Optimization | SUT
Statistical Learning | SUT
Machine Learning | SUT

Skills

Languages: C/C++, Python, Java, Matlab, R, SQL
Libraries: Pandas, NumPy, Matplotlib, Keras, Sklearn, MLflow, Simulink, Cvxpy, JAX, Pytorch,
Tools: Git/GitHub, VS Code, Linux, Spice, LaTeX

Languages

Persian(Native)
English(Highly Fluent)[TOEFL R:24 L:28 S:24 W:24] [GRE Q:170]
French(beginner)